

Sayan Karmakar

Software Engineer

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SUMMARY

- Results-driven Software Engineer with strong C++ expertise, specializing in embedded systems, networking protocols, & cross-platform development.
- Delivered high-impact solutions including HTTP stacks for mobile & IoT diagnostics, optimized client systems & managed complex IoT interfaces using cutting-edge technologies. Proficient with Android NDK & network programming with deep expertise in modem software, protocol implementation & performance optimization. Skilled in systems programming, building scalable backend systems, low latency applications, & cross-platform solutions leveraging modern C++ practices.
- Available for relocation across India and EMEA Zones. Eligible for [EU Blue Card](#).

EXPERIENCE

Qualcomm - Software Engineer

January 2023 – Present

Modem Software & Networking Team - QData (Part of the IMS - IP Multimedia Subsystems Team)

- **Development of an HTTP Stack** : Built systems for mobile & IoT device diagnostics over TCP, UDP, or CoAP and developed an Android HTTP stack on HLOS (AOSP) for ML training data/log packets.
- **Android NDK Development** : Development & solving the Android based issues & unit testing for various parts of the framework by integrating the code into various Embedded Devices/Clients.
- **Managed IoT Interfaces** : Worked with C++ on the LwM2M framework to manage the interface between IoT devices, exchanging data from the modem to the server and HLOS using Google Protocol Buffers.
- **Optimized Client Systems & Cross-Platform Development**: Led end-to-end performance optimization, improving power efficiency & reducing latency using advanced debugging & profiling to eliminate critical bottlenecks. Enhanced power management & memory utilization, and executed stress testing & benchmarking to ensure system reliability. Spearheaded cross-platform code porting across Qualcomm Snapdragon ecosystems (Samsung, OnePlus, Vivo, & Oppo smartphones & Windows laptops), accelerating delivery through strategic resource allocation & cross-functional collaboration.
- **Workflow Improvement** : Utilized Python to handle debugging & automation scripting tasks, significantly improving development workflows in a CI/CD environment.
- **MQTT Stack Maintenance for WWAN Modems** : Maintained and optimized an MQTT stack tailored for WWAN-based modems, ensuring reliable message delivery and efficient communication between embedded clients and cloud services.

Airbus - Associate Software Engineer

July 2022 – January 2023

Department of Avionics Software and System Testing

- Involved in **Software Development Life Cycle (SDLC)**, starting from requirement gathering and performed **Object Oriented Analysis** and Design for development of key interfaces. V Development Life cycle was followed.
- Designed and implemented various classes and methods to increase the performance of the existing software for the **Flight Warning System (FWS)**.
- Created G-sites and custom components for communication and collaboration within Airbus.

TECHNICAL SKILLS

- **Programming Languages**: C/C++, Python, Rust, Java, Javascript, React
- **Libraries/Frameworks**: Embedded Firmware, C++ Sockets, TCP/IP Network Stack, Google Protocol Buffers, Kernel Programming, Android Bootloader, Android NDK, Android Open Source (AOSP), Android Internals, Secure Communication using OpenSSL, Basics of CUDA Programming, QNX/Yocto, Bitbake Recipes.
- **Cross Platform**: CMake, GDB, LLVM, LLDB, Visual Studio, Scons, Valgrind, Bullseye, GCov & other C++ build systems.
- **Networking Protocols**: HTTPS, UDP, CoAP, DTLS, TCP/IP, MQTT, LTE/NR
- **Code Management**: Git, Github, Gerrit, Perforce
- **Networking Tools** : Wireshark, TCPDump
- **Operating Systems**: Windows x64, Unix, Linux, Windows on ARM

EDUCATION

Jadavpur University

2018-2022

B.E.(Hons) in Instrumentation and Electronics Engineering

- CGPA - 8.36 / 10

RESEARCH PAPERS

1. Low-cost air pollution monitoring device based on AQI (air quality index) | [🔗 Link](#)
2. Development of an IoT based robust architecture for environmental monitoring using UAV | [🔗 Link](#)